

CTMT Global Similarity Atlas

Orbit–Stabilizer Classification, Chart Atlases, and Observable Resolution Types

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Abstract

Chronotopic Metric Theory (CTMT) assigns to each observable transport experiment a local resolution geometry

$$\mathcal{G} = (O; R, N, \Sigma),$$

where $R = \text{Im}(J)$ is the resolved sector induced by a response operator, $N = \text{Ker}(J^T)$ is the unresolved sector, and Σ is empirical covariance. Previous foundational layers define admissible morphisms, collapse, and holonomy. This paper develops the next global layer: the *Global Similarity Atlas*. The atlas classifies observable states, charts, and local resolution geometries by their orbits under admissible automorphisms and by compatibility of these orbits on overlaps.

The central construction is an atlas whose charts are not merely coordinate patches, but local similarity classifications of $\mathcal{G} = (O; R, N, \Sigma)$. Each chart carries an automorphism group, stabilizer groups, similarity orbits, covariance block type, coupling type, and holonomy data. Transition maps between charts are admissible exactly when they preserve the resolution type. The main result is an atlas lock: a global similarity atlas exists precisely when local resolution types can be glued by admissible transitions with consistent orbit–stabilizer data and bounded holonomy. Failure of this gluing is not a numerical inconvenience; it is a specific global obstruction measured by rank jumps, stabilizer discontinuity, sector mismatch, covariance-type mismatch, coupling incompatibility, or holonomy-level collapse.

The paper is written as a foundational CTMT document. It gives definitions, theorems, proof sketches, counterexamples, a computational protocol, a master summary table, non-claims, limitations, and reproducibility appendices. The intent is operational: the atlas is a classification device for observable transport geometry, not a claim that a particular physical system has been completely described.

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1 Introduction

Local CTMT begins with a single observable reconstruction. Given a local response operator J and empirical covariance Σ , one constructs

$$R = \text{Im}(J), \quad N = \text{Ker}(J^T), \quad \mathcal{G} = (O; R, N, \Sigma).$$

This local object already determines rank, resolved and unresolved projectors, covariance blocks, Fisher-type response geometry, null covariance, coupling, and the admissible automorphism group. However, a single chart is not enough for global transport. If the same system is observed across several operating regions, sensor layouts, bandwidths, or perturbation regimes, CTMT must compare many local resolution geometries.

The present paper introduces the *Global Similarity Atlas*. The word “similarity” is used in the group-action sense. Two observable states are similar if they lie in the same orbit under the automorphism group preserving resolution geometry. A global atlas is then a compatible collection of local similarity classifications glued by admissible transitions. In this language, CTMT globalization becomes a classification problem:

Can local observable resolution types be glued into a coherent global atlas of similarity classes?

This question is sharper than asking whether a metric exists globally. Metrics, connections, Fisher matrices, covariance blocks, winding indices, and holonomy are treated as derived structures. The atlas records the primitive observable classification from which those structures arise.

1.1 Main contributions

This paper contributes the following foundational pieces.

1. Definition of local similarity type for $\mathcal{G} = (O; R, N, \Sigma)$.
2. Orbit–stabilizer classification of observable states under $\text{Aut}(\mathcal{G})$.
3. Definition of a similarity chart and global similarity atlas.
4. Transition compatibility conditions on overlaps.
5. Atlas gluing theorem and obstruction classification.
6. Relationship between atlas consistency, holonomy, and collapse.
7. Computational protocol for building an atlas from finite data.
8. Counterexamples showing why rank-only atlases fail.

2 Local Similarity Type

Definition 2.1 (Resolution geometry). A local CTMT resolution geometry is a quadruple

$$\mathcal{G} = (O; R, N, \Sigma),$$

where $O \cong \mathbb{R}^m$, $R \subset O$, $N \subset O$, $O = R \oplus N$, and Σ is an empirical covariance operator on O .

Definition 2.2 (Adapted covariance type). Let $Q = [U_R \ U_N]$ be an orthonormal basis adapted to $O = R \oplus N$. The adapted covariance type of $\mathcal{G}$ is the block matrix

$$Q^T \Sigma Q = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^T & \Sigma_N \end{pmatrix}.$$

The block C_{RN} is called the resolved–null coupling type.

Definition 2.3 (Local similarity type). The local similarity type of $\mathcal{G}$ is the tuple

$$\tau(\mathcal{G}) = (\dim O, \dim R, \dim N, [\Sigma_R], [\Sigma_N], [C_{RN}], \text{Aut}(\mathcal{G})),$$

where brackets denote equivalence under adapted orthogonal changes of basis preserving R and N .

Remark 2.4. The similarity type is deliberately not just rank. Rank only records $\dim R$. It forgets the null sector, covariance scale, coupling, stabilizers, and holonomy-relevant automorphisms.

Definition 2.5 (Automorphism group). The automorphism group of a local resolution geometry is

$$\text{Aut}(\mathcal{G}) = \{A \in \text{GL}(O) : A(R) = R, A(N) = N, A^T \Sigma A = \Sigma\}.$$

3 Observable Similarity Orbits

Definition 3.1 (Similarity of observable states). Let $x, y \in O$. The states x and y are CTMT-similar in the chart $\mathcal{G}$ if there exists $A \in \text{Aut}(\mathcal{G})$ such that

$$y = Ax.$$

Definition 3.2 (Similarity orbit). The similarity orbit of $x \in O$ is

$$\text{Orb}_{\mathcal{G}}(x) = \{Ax : A \in \text{Aut}(\mathcal{G})\}.$$

Definition 3.3 (Stabilizer). The stabilizer of $x \in O$ is

$$\text{Stab}_{\mathcal{G}}(x) = \{A \in \text{Aut}(\mathcal{G}) : Ax = x\}.$$

Theorem 3.4 (Orbit–stabilizer structure). Assume $\text{Aut}(\mathcal{G})$ acts smoothly on O . Then each similarity orbit is isomorphic, as a homogeneous space, to

$$\text{Orb}_{\mathcal{G}}(x) \cong \text{Aut}(\mathcal{G}) / \text{Stab}_{\mathcal{G}}(x).$$

Proof. The map $\text{Aut}(\mathcal{G}) \rightarrow \text{Orb}_{\mathcal{G}}(x)$ given by $A \mapsto Ax$ is surjective by definition. Two automorphisms give the same orbit point exactly when they differ by an element stabilizing x . Therefore the orbit is the quotient by the stabilizer. $\square$

Corollary 3.5 (Stabilizer records local degeneracy). Large stabilizers correspond to high residual symmetry of the observable state; small stabilizers correspond to more rigid states. Stabilizer change is therefore an atlas-level diagnostic of changing observable resolution type.

4 Similarity Charts

Definition 4.1 (Similarity chart). A CTMT similarity chart is a tuple

$$\mathfrak{U}_{\alpha} = (U_{\alpha}, \mathcal{G}_{\alpha}, \text{Aut}(\mathcal{G}_{\alpha}), \{\text{Orb}_{\alpha}(x)\}_{x \in U_{\alpha}}, \{\text{Stab}_{\alpha}(x)\}_{x \in U_{\alpha}}),$$

where U_{α} is an observable neighborhood and $\mathcal{G}_{\alpha}$ is the local resolution geometry reconstructed on it.

Definition 4.2 (Chart fingerprint). The computational fingerprint of a chart is

$$\mathfrak{f}_{\alpha} = (r_{\alpha}, n_{\alpha}, \text{spec } \Sigma_{R,\alpha}, \text{spec } \Sigma_{N,\alpha}, \|C_{RN,\alpha}\|, \dim \text{Aut}(\mathcal{G}_{\alpha}), \dim \text{Stab}_{\alpha}(x), \mathcal{H}_{\alpha}),$$

where $r_{\alpha} = \dim R_{\alpha}$, $n_{\alpha} = \dim N_{\alpha}$, and $\mathcal{H}_{\alpha}$ denotes selected holonomy data based at the chart.

Remark 4.3. The fingerprint is not the theory. It is a finite numerical summary used to compare charts in computation. The theory is the equivalence class of resolution geometries and their orbit–stabilizer structures.

5 Atlas Transitions

Definition 5.1 (Similarity transition). A transition map between similarity charts

$$\Phi_{\alpha\beta} : \mathfrak{U}_{\beta} \rightarrow \mathfrak{U}_{\alpha}$$

is admissible if its tangent map preserves resolution geometry:

$$D\Phi_{\alpha\beta}(R_{\beta}) = R_{\alpha}, \quad D\Phi_{\alpha\beta}(N_{\beta}) = N_{\alpha}, \quad D\Phi_{\alpha\beta}^T \Sigma_{\alpha} D\Phi_{\alpha\beta} = \Sigma_{\beta}.$$

It is similarity-compatible if, in addition, it maps orbits to orbits and stabilizers by conjugation:

$$\begin{aligned} \Phi_{\alpha\beta}(\text{Orb}_{\beta}(x)) &= \text{Orb}_{\alpha}(\Phi_{\alpha\beta}x), \\ \text{Stab}_{\alpha}(\Phi_{\alpha\beta}x) &= D\Phi_{\alpha\beta} \text{Stab}_{\beta}(x) D\Phi_{\alpha\beta}^{-1}. \end{aligned}$$

Proposition 5.2 (Admissibility implies orbit compatibility). *If $\Phi_{\alpha\beta}$ is an admissible isomorphism of resolution geometries, then it induces an isomorphism of automorphism groups by conjugation and maps similarity orbits to similarity orbits.*

Proof. Let $A \in \text{Aut}(\mathcal{G}_\beta)$. Since $\Phi_{\alpha\beta}$ preserves the resolution geometry, the conjugate $D\Phi_{\alpha\beta}AD\Phi_{\alpha\beta}^{-1}$ preserves $\mathcal{G}_\alpha$. Thus conjugation maps automorphisms to automorphisms. If $y = Ax$, then $\Phi_{\alpha\beta}(y) = (D\Phi_{\alpha\beta}AD\Phi_{\alpha\beta}^{-1})\Phi_{\alpha\beta}(x)$, so orbits map to orbits. Stabilizers transform by the same conjugation. $\square$

6 Global Similarity Atlas

Definition 6.1 (Global similarity atlas). A global CTMT similarity atlas is a collection

$$\mathcal{A} = \{\mathfrak{U}_\alpha, \Phi_{\alpha\beta}\}_{\alpha, \beta \in I}$$

such that:

- (a) the neighborhoods U_α cover the observed domain;
- (b) each $\mathfrak{U}_\alpha$ is a similarity chart;
- (c) each nonempty overlap carries an admissible similarity transition;
- (d) on triple overlaps the cocycle condition holds:

$$\Phi_{\alpha\beta} \circ \Phi_{\beta\gamma} = \Phi_{\alpha\gamma};$$

- (e) loop products remain bounded admissible holonomies.

Definition 6.2 (Atlas equivalence). Two global similarity atlases are equivalent if they admit a common refinement whose transition maps preserve local similarity type, orbit–stabilizer data, covariance block type, and holonomy class.

7 Atlas Gluing Theorem

Theorem 7.1 (Global similarity atlas gluing). *Let $\{\mathfrak{U}_\alpha\}_{\alpha \in I}$ be a finite family of local similarity charts. A global similarity atlas exists if the following conditions hold:*

- (i) equal rank and null dimensions on overlaps;
- (ii) admissible sector transitions $R_\beta \rightarrow R_\alpha$ and $N_\beta \rightarrow N_\alpha$;
- (iii) covariance compatibility $D\Phi_{\alpha\beta}^T \Sigma_\alpha D\Phi_{\alpha\beta} = \Sigma_\beta$;
- (iv) coupling compatibility of C_{RN} under adapted block transport;
- (v) automorphism groups conjugate on overlaps;
- (vi) stabilizers conjugate along orbit representatives;
- (vii) cocycle consistency on triple overlaps;
- (viii) bounded admissible holonomy around tested loops.

Conversely, failure of any required condition is an atlas obstruction.

Proof. The listed conditions provide exactly the data needed to identify local similarity charts on overlaps. Conditions (i)–(iv) glue the primitive resolution geometries. Conditions (v)–(vi) glue the orbit–stabilizer classification. Condition (vii) ensures consistency of transition composition. Condition (viii) prevents loop products from leaving the admissible class or becoming unbounded. Therefore the charts assemble into a global atlas. If any condition fails, the corresponding structure cannot be glued, producing a specific obstruction. $\square$

Corollary 7.2 (Rank-only classification is insufficient). *Equal rank on all overlaps is necessary but not sufficient for a global similarity atlas.*

Proof. Equal rank gives only condition (i). The theorem requires sector, covariance, coupling, automorphism, stabilizer, cocycle, and holonomy compatibility. Any of these may fail while rank remains constant. $\square$

8 Atlas Obstructions

Definition 8.1 (Atlas obstruction). An atlas obstruction is any measurable failure preventing local similarity charts from gluing into a global atlas.

The principal obstruction classes are:

- O1. Rank obstruction:** $\text{rank } J_\alpha \neq \text{rank } J_\beta$ on overlap.
- O2. Sector obstruction:** no admissible map sends $R_\beta \rightarrow R_\alpha$ and $N_\beta \rightarrow N_\alpha$.
- O3. Projector obstruction:** $\|P_{R,\alpha} - TP_{R,\beta}T^{-1}\|$ or $\|P_{N,\alpha} - TP_{N,\beta}T^{-1}\|$ is too large.
- O4. Covariance obstruction:** covariance pullback error exceeds tolerance.
- O5. Coupling obstruction:** C_{RN} cannot be matched under adapted basis transport.
- O6. Automorphism obstruction:** automorphism groups are not conjugate.
- O7. Stabilizer obstruction:** orbit stabilizers jump or fail to conjugate.
- O8. Cocycle obstruction:** transition composition fails on triple overlaps.
- O9. Holonomy obstruction:** loop product leaves admissible class or exceeds conditioning bounds.

Principle 8.2 (Obstruction specificity). *A failed atlas computation should report the first violated obstruction class, not merely state that globalization failed.*

9 Similarity Atlas, Holonomy, and Collapse

Theorem 9.1 (Atlas–holonomy relation). *In a valid global similarity atlas, every closed loop of transitions produces a holonomy element lying in the automorphism group of the base chart:*

$$H_\gamma \in \text{Aut}(\mathcal{G}_{\alpha_0}).$$

Non-identity H_γ is permitted.

Proof. Each transition is admissible and similarity-compatible. The composition of admissible maps is admissible. A closed chain returns to the base chart, hence the loop product is an automorphism of the base resolution geometry. It need not equal the identity. $\square$

Theorem 9.2 (Atlas collapse criterion). *A finite chart family undergoes atlas-level CTMT collapse if and only if no bounded admissible similarity atlas exists for the specified overlap graph and tolerance vector.*

Proof. If a bounded admissible atlas exists, every required overlap, orbit, stabilizer, and holonomy condition is satisfied. Therefore no atlas-level collapse is reported. Conversely, if no such atlas exists, at least one required gluing or holonomy condition fails. This is precisely atlas-level collapse. $\square$

10 Covariance Coupling and Atlas Refinement

Proposition 10.1 (Coupling refines atlas strata). *Suppose two charts have equal rank and equal spectra of Σ_R and Σ_N . If their coupling blocks C_{RN} are not equivalent under adapted sector transformations, then the charts belong to different similarity strata.*

Proof. The coupling block is part of the local similarity type. Adapted sector transformations act on it by left and right multiplication. If no such action matches the two coupling blocks, the covariance operators are not equivalent as resolution geometries, even though their diagonal block spectra agree. $\square$

Corollary 10.2 (Open-object signature). *Nonzero C_{RN} is an atlas-refining observable signature. It does not prove a physical leakage mechanism, but it prevents classification by resolved data alone.*

11 Computational Construction of a Global Similarity Atlas

A finite-data atlas computation proceeds as follows.

- Step 1.** Partition observations into local neighborhoods U_α .
- Step 2.** In each chart compute J_α .
- Step 3.** Compute R_α , N_α , $P_{R,\alpha}$, $P_{N,\alpha}$ by SVD.
- Step 4.** Estimate Σ_α from repeated observations.
- Step 5.** Lift covariance into the adapted basis and extract $\Sigma_{R,\alpha}$, $C_{RN,\alpha}$, and $\Sigma_{N,\alpha}$.
- Step 6.** Compute chart fingerprints f_α .
- Step 7.** On overlaps solve for transition candidates $T_{\alpha\beta}$ minimizing sector, covariance, and coupling defects.
- Step 8.** Test automorphism group conjugacy and stabilizer compatibility.
- Step 9.** Test triple-overlap cocycles.
- Step 10.** Compute loop holonomies and condition numbers.
- Step 11.** Emit pass/fail atlas report and obstruction class.

12 Master Atlas Table

Table 1: Global Similarity Atlas: one-dataset reconstruction summary.

Object	Computation	Atlas role
Rank	SVD of J_α	primary stratum dimension
Resolved sector	$U_{R,\alpha}$	observable response subspace
Null sector	$U_{N,\alpha}$	unresolved observable subspace
Projectors	P_R, P_N	sector comparison on overlaps
Covariance	empirical Σ_α	observation resolution type
Coupling	$C_{RN,\alpha}$	atlas refinement by leakage/coupling
Automorphism group	preservers of R, N, Σ	local symmetry type
Orbit	$\text{Aut}(\mathcal{G})x$	similarity class of state
Stabilizer	fixed subgroup of x	residual local degeneracy
Transition	pullback preserving $\mathcal{G}$	gluing map
Cocycle	triple-overlap composition	atlas consistency
Holonomy	loop product	global obstruction/trivialization defect

13 Counterexamples

Example 13.1 (Equal rank, incompatible covariance). Let two charts have identical R and N but covariance blocks

$$\Sigma_{R,1} = I, \quad \Sigma_{R,2} = \text{diag}(1, 2).$$

Rank and sectors agree, but covariance type differs. A rank-only atlas would glue them; a CTMT similarity atlas rejects the gluing unless a permitted transition pulls one covariance to the other.

Example 13.2 (Equal covariance spectra, incompatible coupling). Two charts may have identical spectra of Σ_R and Σ_N but different coupling blocks C_{RN} . If the coupling blocks are not equivalent under sector-preserving transformations, the charts lie in different atlas strata.

Example 13.3 (Orbit jump). A chart family may preserve rank, sectors, and covariance dimensions, while the stabilizer dimension of a distinguished observation changes. This indicates an orbit-type transition and must be recorded as an atlas stratum boundary.

Example 13.4 (Cocycle failure). Transitions may be pairwise admissible on overlaps but fail

$$\Phi_{\alpha\beta}\Phi_{\beta\gamma} = \Phi_{\alpha\gamma}$$

on a triple overlap. Then no global atlas exists for that cover without refinement.

14 Non-Claims and Limitations

1. The Global Similarity Atlas is an observable classification device, not a claim of complete physical ontology.
2. Equal atlas type does not prove two physical systems are identical. It means they are indistinguishable by the chosen observable resolution geometry.
3. Atlas failure does not automatically imply physical singularity. It means resolution geometries could not be glued under the stated protocol.

4. Holonomy is not automatically collapse. Non-trivial holonomy is allowed when it remains admissible.
5. Coupling C_{RN} is not assumed to be a physical force or causal leak. It is an observable covariance block that refines similarity classification.
6. Stabilizer dimension is sensitive to degeneracies and finite-sample estimation; statistical tolerance must be stated explicitly.

15 Open Problems

Open Problem 15.1 (Similarity strata classification). Classify all local similarity types of $(O; R, N, \Sigma)$ for fixed $(\dim O, \dim R, \dim N)$ up to adapted sector transformations.

Open Problem 15.2 (Coupling normal forms). Find canonical normal forms for C_{RN} under the action of covariance-preserving sector automorphisms.

Open Problem 15.3 (Statistical atlas stability). Derive finite-sample confidence bounds for rank, projector distance, covariance block error, stabilizer dimension, and holonomy defect.

Open Problem 15.4 (Atlas refinement algorithms). Develop algorithms that automatically refine a cover until cocycle and holonomy errors either close or yield a minimal obstruction certificate.

Open Problem 15.5 (Physical realization). Identify physical observable systems for which atlas strata correspond to known transport regimes, phase transitions, or sensor-resolution changes.

16 Conclusion

The Global Similarity Atlas is the CTMT mechanism for moving from local resolution geometry to global observable classification. A single chart tells us what is resolved, unresolved, covariant, coupled, and symmetric at one local experiment. An atlas tells us whether those local classifications can be glued across an observed domain.

The key conclusion is that global similarity is stronger than rank stability. It requires compatible sectors, covariance blocks, coupling, automorphism groups, stabilizers, cocycles, and holonomy. When these structures glue, CTMT obtains a coherent global observable classification. When they fail, the failure is not vague: it appears as a named atlas obstruction. This makes the Global Similarity Atlas a natural bridge between local CTMT reconstruction, holonomy, collapse, and future physical interpretation.

A Appendix A: Atlas JSON Schema

A reproducible atlas computation should emit the following fields.

```
{
  "charts": [
    {
      "chart_id": "U0",
      "J": [[...]],
      "rank": 4,
      "U_R": [[...]],
      "U_N": [[...]],
    }
  ]
}
```

```

    "P_R": [...],
    "P_N": [...],
    "Sigma": [...],
    "Sigma_block": [...],
    "Sigma_R": [...],
    "C_RN": [...],
    "Sigma_N": [...],
    "automorphism_summary": {...},
    "orbit_summary": {...},
    "stabilizer_summary": {...},
    "fingerprint": {...}
  }
],
"overlaps": [
  {
    "from": "U1",
    "to": "U0",
    "transition": [...],
    "rank_error": 0,
    "sector_error": 1.0e-12,
    "covariance_error": 2.0e-11,
    "coupling_error": 3.0e-11,
    "stabilizer_error": 0,
    "pass": true
  }
],
"cocycles": [...],
"holonomies": [...],
"atlas_obstructions": []
}

```

B Appendix B: Recommended Numerical Health Bars

Table 2: Recommended health bars for atlas construction.

Health bar	Quantity	Failure mode
Rank	$ r_\alpha - r_\beta $	rank obstruction
Resolved sector	$\ P_{R,\alpha} - TP_{R,\beta}T^{-1}\ $	sector obstruction
Null sector	$\ P_{N,\alpha} - TP_{N,\beta}T^{-1}\ $	sector obstruction
Covariance	$\ T^T \Sigma_\alpha T - \Sigma_\beta\ $	covariance obstruction
Coupling	$\ C_{RN,\alpha}(T) - C_{RN,\beta}\ $	coupling obstruction
Stabilizer	dimension/conjugacy error	orbit-type jump
Cocycle	$\ T_{\alpha\beta}T_{\beta\gamma} - T_{\alpha\gamma}\ $	atlas obstruction
Holonomy	$\text{dist}(H_\gamma, \text{Aut}(\mathcal{G}))$	holonomy collapse
Conditioning	$\kappa(H_\gamma)$	transport blow-up

C Appendix C: Minimal Python Pseudocode

```

for chart in charts:
    J = estimate_jacobian(chart.data)

```

```

U, s, Vt = svd(J, full_matrices=True)
rank = count(s > tol)
U_R = U[:, :rank]
U_N = U[:, rank:]
P_R = U_R @ U_R.T
P_N = U_N @ U_N.T
Sigma = empirical_covariance(chart.residuals) + ridge * I
Q = concatenate([U_R, U_N], axis=1)
Sigma_block = Q.T @ Sigma @ Q
fingerprint = build_fingerprint(rank, Sigma_block)

for overlap in overlaps:
    T = solve_transition(chart_a, chart_b)
    errors = compute_atlas_errors(T, chart_a, chart_b)
    report_pass_fail(errors)

for triple in triples:
    cocycle_error = norm(T_ab @ T_bc - T_ac)

for loop in loops:
    H = product(loop.transitions)
    holonomy_error = distance_to_automorphism_group(H)
    condition = cond(H)

```

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